



Airflow to the rescue: managing chemical emergencies



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About me



- Product Owner @ MeteoSim
- Industrial Engineer
- Born near Barcelona
- 4th Airflow Summit!



What we do



At MeteoSim, we provide **innovative environmental solutions**

20 +

years of experience

700 +

numerical simulations
run daily

80 +

countries



Our customers



Insurance companies

Assess risk for extreme weather conditions

Oil & Gas companies

Manage the environmental impact of their operations and react to emergencies

Mining companies

Manage the environmental impact of their operations

Waste & water companies

Manage odor complaints

Public institutions

Assess climate change impact



Some of our workloads



Some of our workloads



01

Weather forecasting

Core service Foundation running automated continuous atmospheric models.



02

Chemical Emergencies

Real-time incident intelligence for rapid hazard dispersion management.



03

Custom Projects

Faster task execution with HITL workflows.



04

Sensor Verification

Proprietary automated QA/QC environmental data.



05

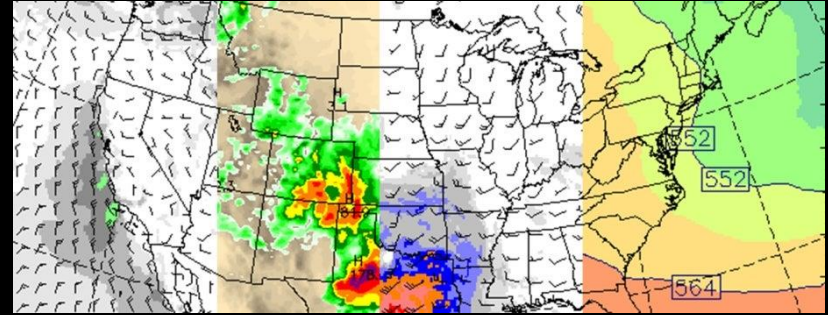
AI Model Training

Scalable pipeline orchestration unlocking new domain opportunities.

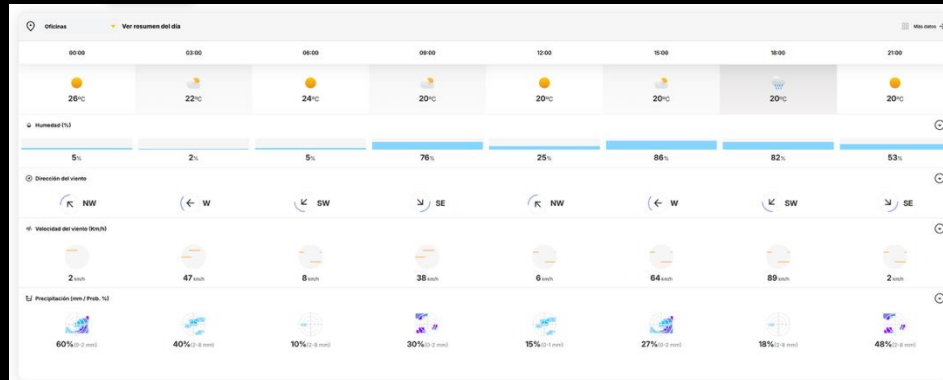
Weather forecasts



- We run 3D local weather simulations
- For the next 24 to 72 hours
- Up to 333 m resolution (1100 ft)
- Up to 10 minutes resolution



Source: ucar.edu



Weather forecasts

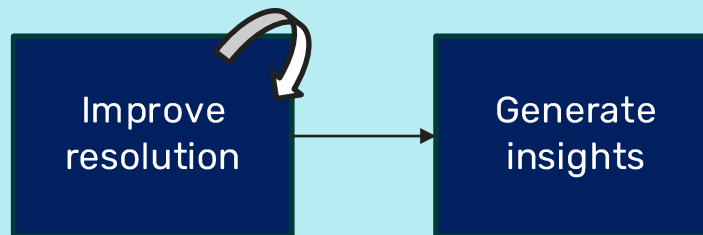
Download global models

1 Dag

Shared for all our local
simulations

Generate local forecasts

1 Dag per site



Chemical emergencies

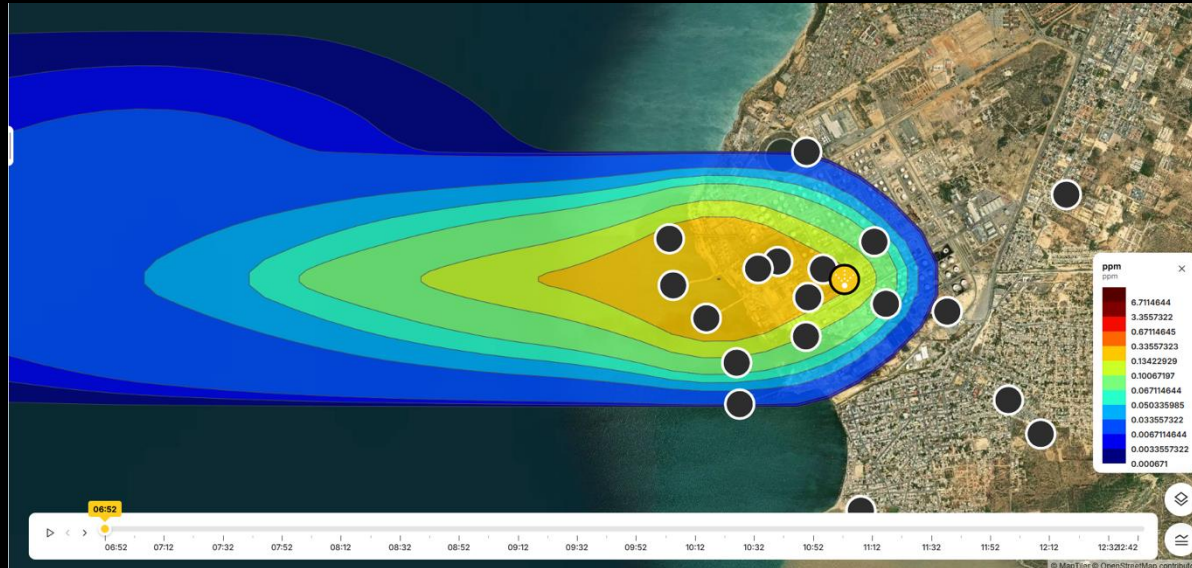


Ona
Chem

AI-generated

Chemical emergencies

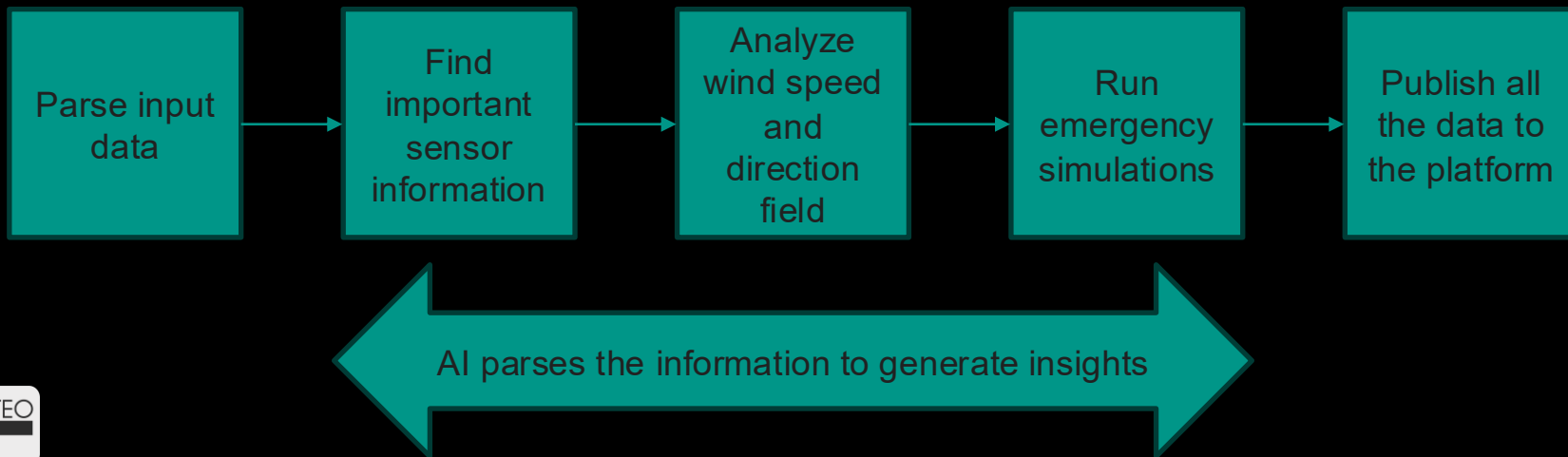
OnaChem React uses a proprietary algorithm to help users manage active chemical emergencies



Chemical emergencies



- The algorithm is based on an Airflow Dag, that decides which simulations to run and generates insights.
- The user can manage the emergency based on a flow with all the analysis results.



Chemical emergencies



The customer never sees Airflow, but their data does:

Up to 4 times a day

We generate a fresh, high resolution 3D weather forecast

In the event of an emergency

An Airflow Dag orchestrates the information generation in less than 2 minutes.

The simulations run in a Celery cluster with high availability

HITL: simulation calibration & consultancy



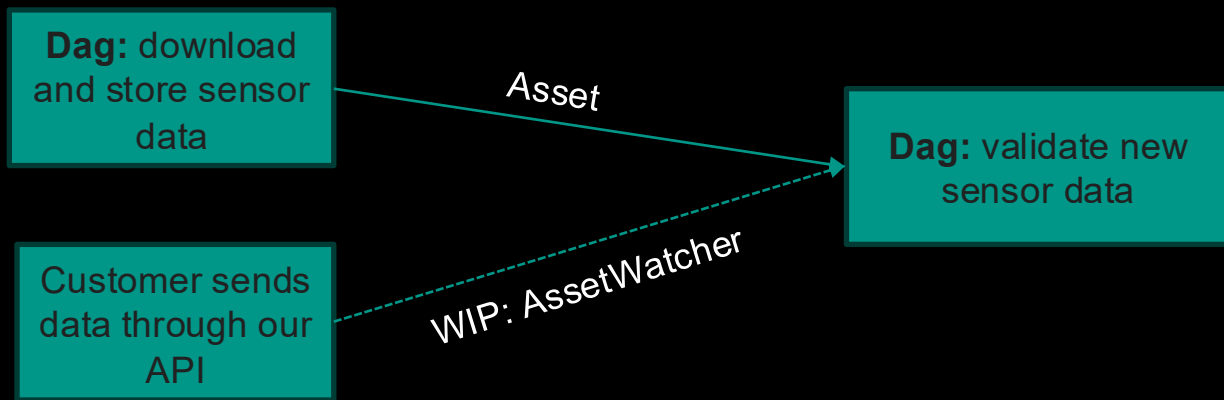
- Weather forecasts require calibration and validation based on real sensor data.
- We have developed several HITL-based Dags to do it.
- The HITL operators ask the necessary questions to properly calibrate and validate simulations.

87% *time saved*

A screenshot of the Apache Airflow web interface showing the configuration for a DAG named "Trigger Dag - solutions.validate_prec_fcst". The interface is in a dark theme. At the top, there are two tabs: "Single Run" (selected) and "Backfill". Below the tabs, the "Logical Date" is set to "08/27/2026, 03:26:55 PM". The main section is titled "Info general del projecte" and contains several form fields: "Codi comercial del projecte" with the value "SHA25363A.EMI", "Codi de la validació" with the value "EXP1_oct2025-feb2026", "Directorio de sortida" with the value "/Projects/my-project/validation_results", and "Partició de Bright" with the value "low". There is also a toggle switch for "Executa amb el meu propi usuari de Bright" which is currently turned on. At the bottom right, there is a "Trigger" button.

Sensor data

- Our platforms can ingest weather, air quality and process data from our customers.
- In some cases, we connect to their API to obtain the data.
- We have automated quality control based on a proprietary algorithm that flags suspicious data for manual verification.



AI training



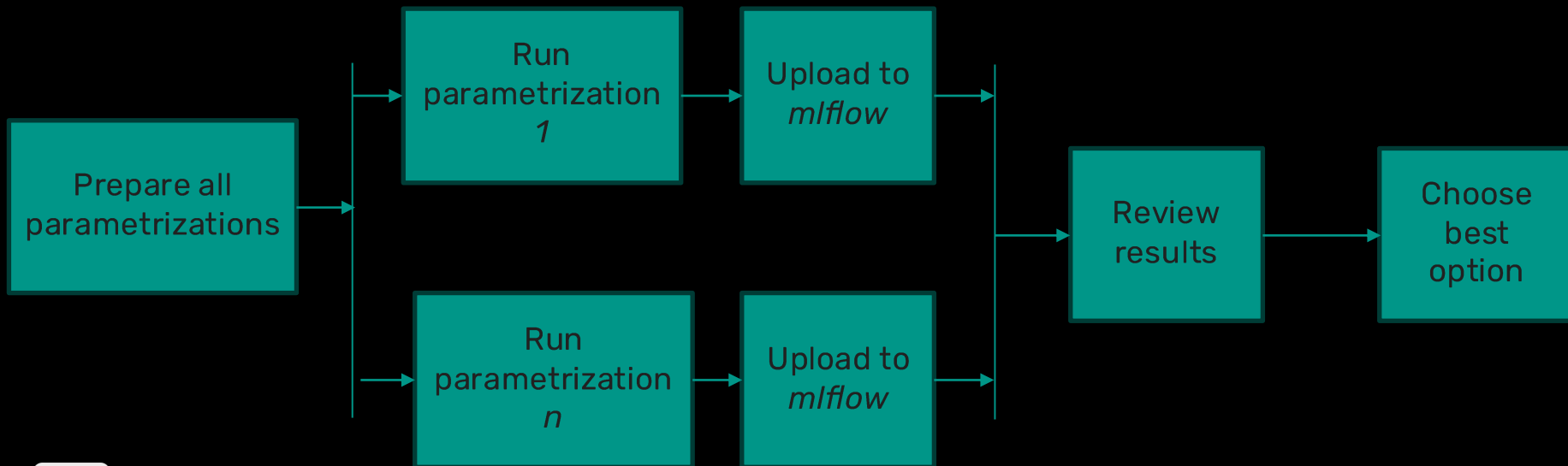
- New research and opportunities in weather, air quality and emergency management
- We offer *nowcasting* services: forecast what's going to happen in real-time for the next few minutes/hours



AI training



- Airflow helps us train AI models for the nowcasting service
- We use mapped tasks and the HITL operator to train our models

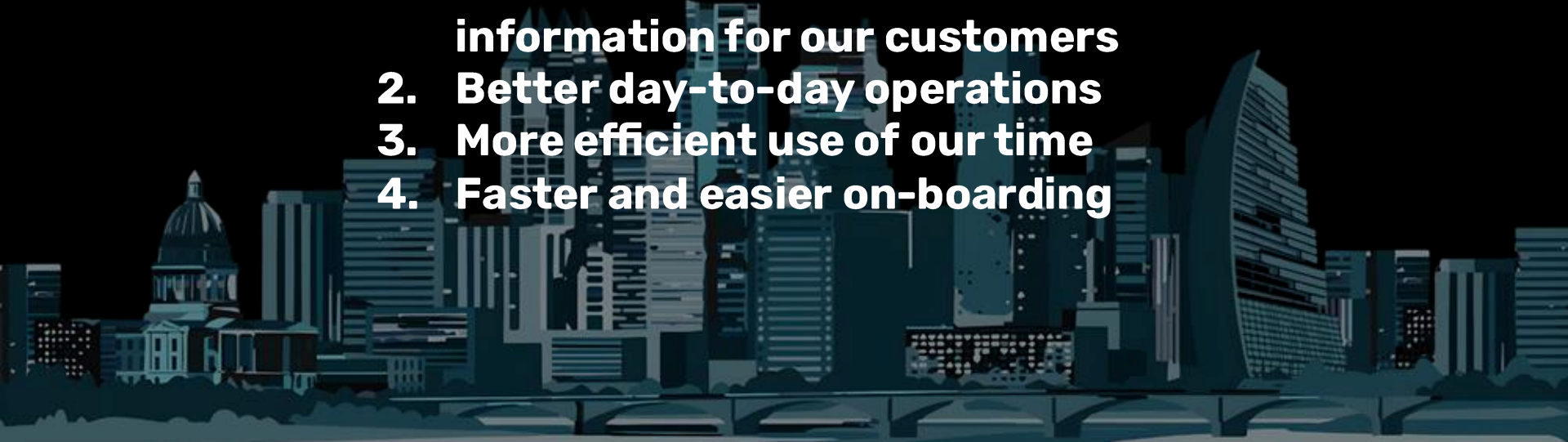


Airflow for MeteoSim



Airflow enables:

- 1. More accurate and reliable information for our customers**
- 2. Better day-to-day operations**
- 3. More efficient use of our time**
- 4. Faster and easier on-boarding**



Questions?



 eloi-codina